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Improved Backscattering Communication Range Using Reflection Amplifiers

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Abstract
As next-generation communication technologies continue to advance, the need for low-power wireless communication is increasing. Among low-power communication technologies, backscatter communication achieves wireless communication energy efficiency on the order of a few picojoules per bit; however, its limited communication range remains a key drawback. This study investigates a method for improving the range of backscatter communication by using a reflection amplifier. An active backscatter tag was designed by integrating a reflection amplifier, a hybrid coupler, and a high-isolation array antenna. Signals were transmitted and received using a software-defined radio (SDR). When the proposed backscatter tag was used at the same location, the magnitude of the backscattered signal increased by approximately 1.5 times. In addition, the experiments confirmed that the communication range of the backscatter tag increased by 40 cm compared with that of a passive tag. The proposed amplified-reflection tag design technique can be applied to ultralow-power indoor short-range Internet of Things (IoT) networks.

Key words: Low-Power Communication; Active Backscatter Communication; Array Antenna; Software-Defined Radio; Internet of Things

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I. Introduction

Among next-generation communication technologies, low-power wireless communication is essential for building Internet of Things (IoT) networks [1]. In particular, among the various low-power communication technologies, backscatter communication has attracted substantial attention because it offers wireless communication energy efficiency on the order of a few pJ/bit [2]. Backscatter communication is used in a wide range of applications, including radio-frequency identification (RFID), low-power sensors, and target tracking. Because it communicates by using the wireless power transmitted from the transmitter, it is vulnerable to propagation attenuation. In addition, at the receiving side, the received RF signal is modulated through load modulation and re-radiated without a separate RF transmitter; therefore, the required wireless communication power is extremely low. Owing to these characteristics, backscatter communication generally has a relatively shorter communication range than other wireless communication technologies [3].

In this paper, to improve the communication range of backscatter communication, the signal-to-noise ratio (SNR) is improved by adopting an active backscatter tag structure based on a reflection amplifier. Because the proposed active backscatter tag amplifies and reflects the received wireless signal, an improvement in communication range can be expected. The amplified reflector can amplify and reflect the received signal with very low power consumption. Moreover, because it provides higher power-amplification gain when the input voltage applied to the amplified reflector is low, it is well suited to backscatter communication. In addition, by using a high-isolation array antenna, the noise coupled into the receiver can be kept relatively low even when the transmit power of the SDR is increased, which in turn improves the communication range.

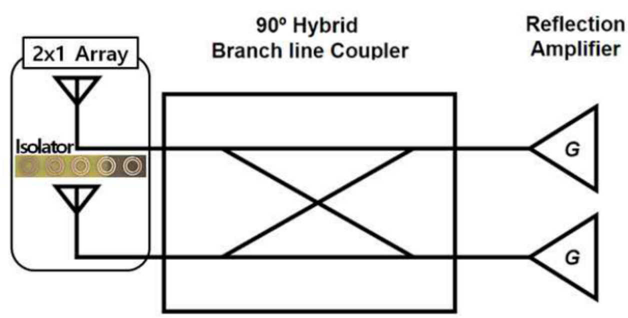


Fig. 1. Block diagram of the backscatter tag.

II. Active Backscatter Tag Design and Measurement

In this study, a backscatter transceiver tag system was designed for the 2.4 GHz industrial, scientific, and medical (ISM) band, which is widely used by Wi-Fi, Bluetooth, and other wireless systems. As shown in Fig. 1, the proposed amplified-reflection active tag consists of a reflection-amplifier circuit, a 90° hybrid coupler, and a 2×1 array antenna. The transmitter and signal-processing chain were implemented using a software-defined radio (SDR).

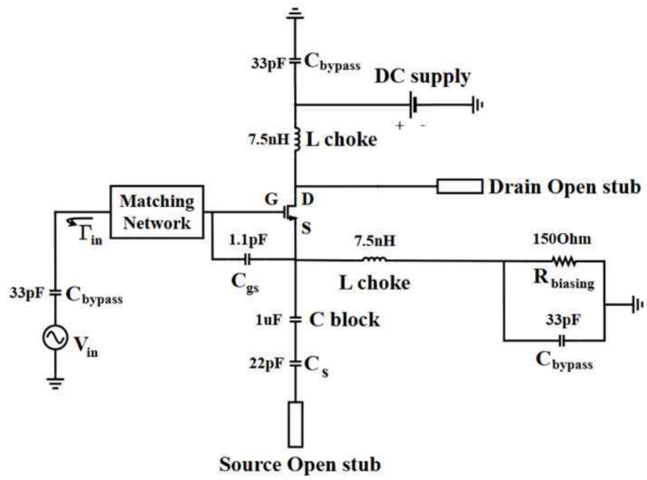


Fig. 2. Proposed schematic of the reflection amplifier.

2-1. Active Reflection-Amplifier Tag Design

As shown in Fig. 2, the reflection amplifier is a one-port active circuit. Therefore, for reflection amplification, the magnitude of the input reflection coefficient must be greater than one ($|\Gamma_{\text{in}}| > 1$):

$$\Gamma_{\text{in}} = \frac{Z_{\text{in}} - Z_0^*}{Z_{\text{in}} + Z_0}. \tag{1}$$

According to (1), for the reflection coefficient to exceed unity, $\text{Re}\{Z_{\text{in}}\}$ must be negative within a range that does not cause oscillation. In this study, a common-source amplifier was designed using a GaAs HJ-FET transistor by connecting open stubs to the source and drain [4]. Simulations were performed using the Advanced Design System (ADS) CAD tool. The frequency band of the drain-connected stub varies with its length, whereas the gain at 2.4 GHz increases as the source-connected stub becomes shorter. The capacitor connected to the source prevents unstable gain variation. To allow component placement, the source-stub length was increased, which reduced the reflection-amplification gain; however, additional gain was obtained by connecting a capacitor between the gate and source. In this manner, the design was carried out by jointly considering the component size and the gain required for amplified reflection.

The designed reflection amplifier was implemented in a coplanar waveguide with ground (CPWG) configuration by placing vias around each microstrip line, which reduced both the substrate thickness and the circuit size. As shown in Fig. 3, the matching network was implemented using a short-stub structure, and the detailed design parameters are summarized in Table 1. The substrate used to design the amplified reflector was 0.6-mm-thick FR-4 ($\varepsilon_r = 4.4$, $\tan\delta = 0.02$), as shown in Fig. 4. The transistor was a NE3509M04 from CEL. GRM capacitors from Murata and 04202HP inductors from Coilcraft were used.

The measurement results of the fabricated reflection amplifier are shown in Fig. 5. The amplifier performance was measured using a Rohde & Schwarz ZNLE6 vector network analyzer. For an input power (P_{in}) of -40 dBm, the designed amplifier exhibited a gain of 6.6 dB at 2.4 GHz when a bias voltage of 1.5 V was applied. Under this condition, the power consumption was 15 mW. The difference between the measured and simulated values was mainly attributed to an inaccurate transistor model. In this work, the basic device model provided by the manufacturer was used, and errors occurred because package information was unavailable.

The 90° hybrid coupler was designed to satisfy $|S_{21}|, |S_{31}| > -4$ dB and $|S_{11}|, |S_{41}| < -25$ dB in the 2.4 GHz band. To reduce the size, the coupler was designed

Table 1. Design parameters of the reflection amplifier (unit: mm).

W_1	0.78	L_1	5.4	L_5	2.1
W_2	0.5	L_2	5.07	L_6	1.2
W_3	0.58	L_3	4.6	L_7	2.5
L_4	10.43	L_8	1.3		

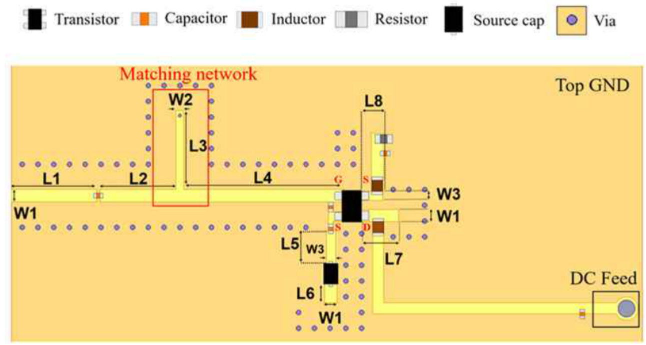


Fig. 3. Reflection-amplifier layout.

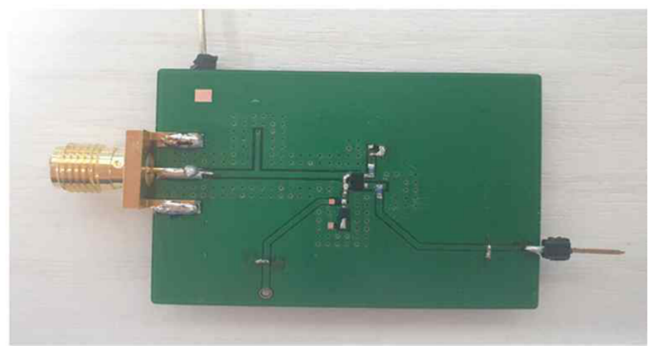


Fig. 4. Fabricated reflection amplifier.

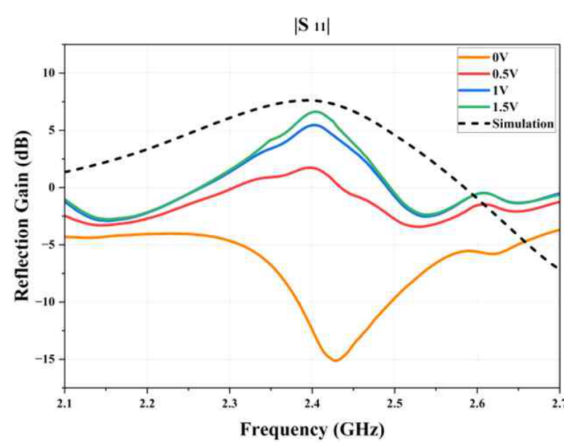


Fig. 5. Measured reflection amplifier: $|S_{11}|$.

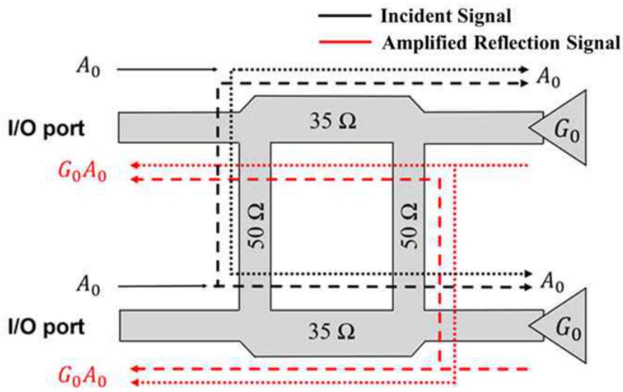


Fig. 6. Block diagram of the coupler and reflection amplifier.

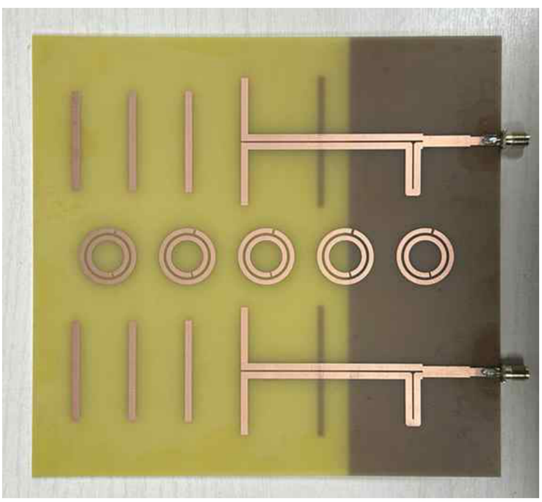


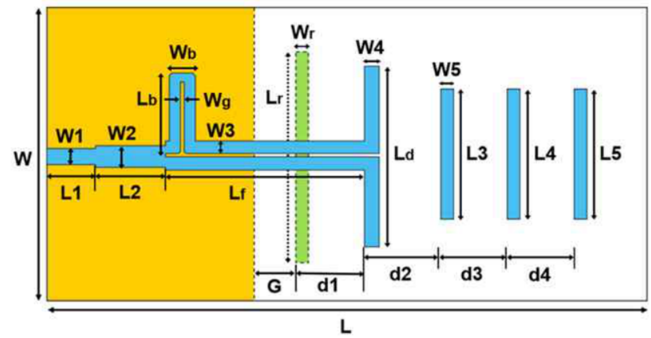
Fig. 7. Proposed 2×1 Yagi-Uda antenna array.

in the basic form of a 90° hybrid coupler. The measured values at 2.4 GHz were -3.5 dB and -4 dB for S_{21} and S_{31}, respectively, and -25 dB and -30 dB for $|S_{11}|$ and $|S_{41}|$, respectively, satisfying the intended design. As shown in Fig. 6, owing to the characteristics of the 90° hybrid coupler, a power gain of G_0 can be obtained at each input/output (I/O) port [5].

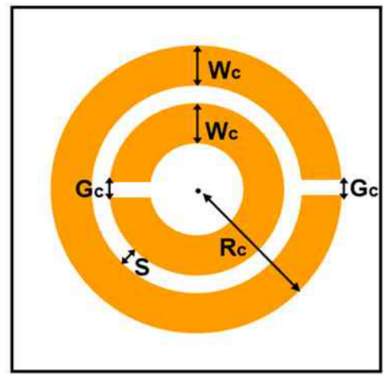
2-2. Design of the 2×1 Yagi-Uda Antenna Array

For the proposed active tag, a two-port 2×1 Yagi-Uda antenna array was designed for connection to the coupler integrated with the reflection amplifiers. A planar Yagi-Uda antenna is appropriate for the proposed active tag because it is easy to fabricate, has a thin and lightweight structure, and provides relatively high antenna gain [6]. In a 2×1 array antenna fabricated on a single substrate, two antennas are formed on the same board, which makes mutual electrical coupling likely and can lead to signal interference. Because this coupling can degrade the performance of the reflection amplifier, suppressing the electrical coupling between antennas is very important. Although increasing the spacing between antennas is one possible solution, a split-ring resonator (SRR) was placed between the antennas, as shown in Fig. 7, to improve isolation while miniaturizing the active tag.

The proposed antenna array was fabricated on a 2-mm-thick



(a) Structure



(b) SRR

Fig. 8. Proposed Yagi-Uda antenna.

FR-4 substrate. Fig. 8 shows the detailed antenna and resonator structures, and the design variables are summarized in Table 2. After the performance of a single antenna, including its radiation pattern and gain, was analyzed, two antennas were arranged with a half-wavelength spacing according to array-antenna theory. The resonator used to suppress mutual coupling between the antennas has the SRR structure shown in Fig. 8(b), and it was designed to resonate at 2.4 GHz. Fig. 9 shows the frequency response of the designed resonator. Based on this response, the resonator was designed to provide appropriate reflection coefficient ($|S_{11}|$) and coupling coefficient ($|S_{21}|$) characteristics at the operating frequency of 2.39–2.41 GHz. In the operating band, the resonator array suppresses both the transmission and reflection of the incident plane wave. The resonator was placed at the center between the transmit and receive antennas of the active tag to improve antenna isolation.

2-3. Antenna Fabrication and Measurement Results

The fabricated 2×1 Yagi-Uda antenna array was measured in an anechoic chamber, as shown in Fig. 10, to obtain its frequency response, antenna gain, and radiation pattern. The measured frequency response of the antenna is shown in Fig. 11. The measured $|S_{11}|$ was below -10 dB in the operating frequency band, and $|S_{21}|$ was below -30 dB. In particular, at the main operating frequency of 2.4 GHz, the coupling coefficient ($|S_{21}|$) was measured as -45.06 dB. Without the resonator, $|S_{21}|$ was -25.49 dB, confirming that high isolation was successfully achieved using the resonator.

The measured radiation patterns are shown in Fig. 12. The measured maximum gain was approximately 7 dBi, which is

Table 2. Design parameters of the antenna and SRR (unit: mm).

W	76	W_b	5.5	L_3	37	L_r	48
W_1	3	W_g	0.5	L_4	36	d_1	28
W_2	4	W_r	2.5	L_5	36	d_2	20.5
W_3	2.5	L	170	L_b	17	d_3	20.5
W_4	2.5	L_1	11	L_d	46.5	d_4	20.5
W_5	2.5	L_2	17	L_f	63.75	G	8.75
G_c	1	S	0.5	R_c	11.5	W_c	1.5

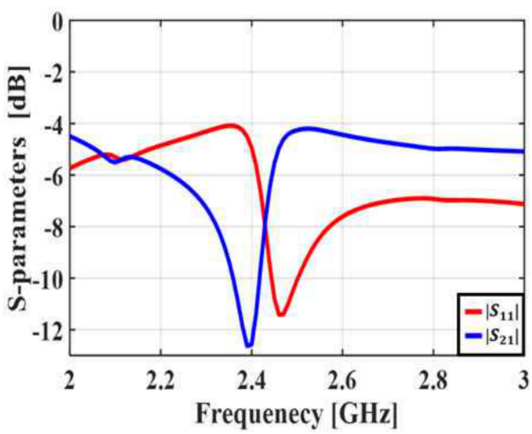


Fig. 9. Frequency response of the designed SRR: $|S_{11}|$ and $|S_{21}|$.

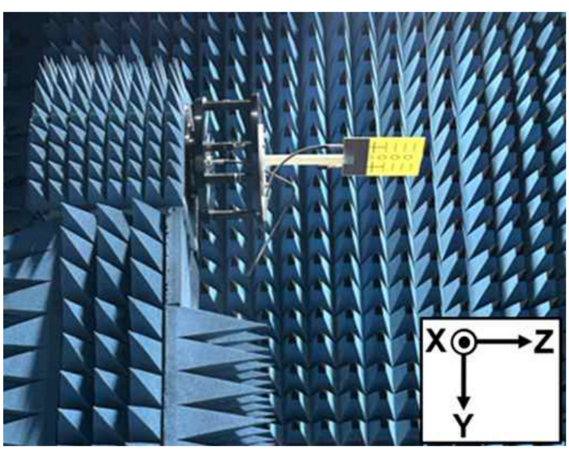


Fig. 10. Antenna measurement environment.

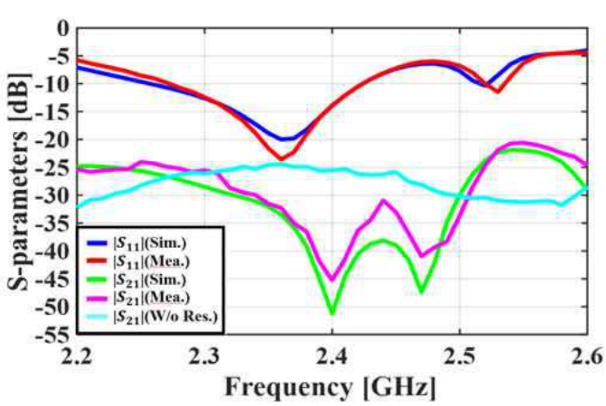
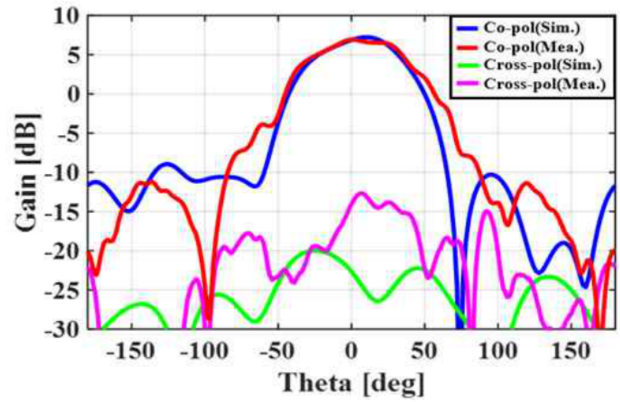
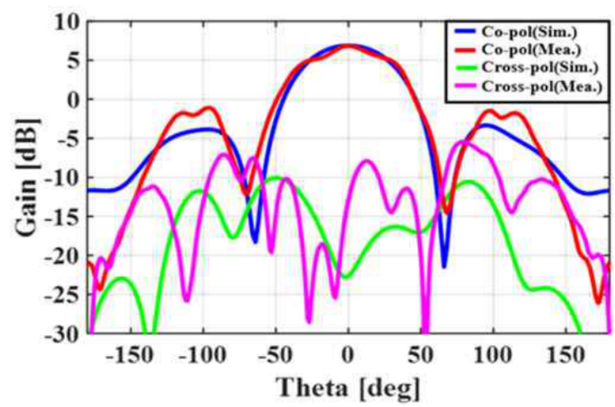


Fig. 11. Measured antenna S-parameters: $|S_{11}|$ and $|S_{21}|$.



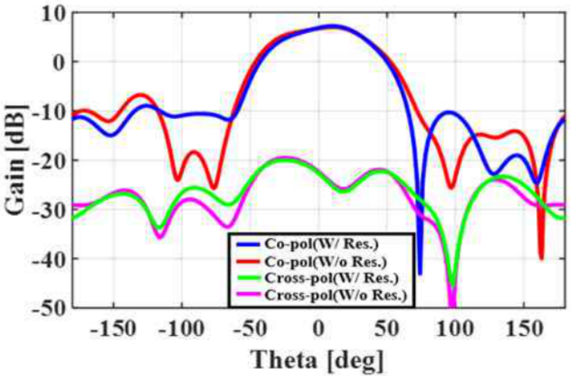
(a) $\phi = 0^\circ$ (xz-plane)



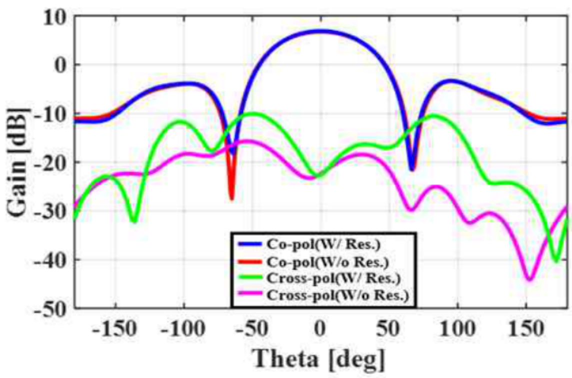
(b) $\phi = 90^\circ$ (yz-plane)

Fig. 12. Measured antenna radiation patterns.

very close to the simulated result of 7.1 dBi. The radiation pattern measured in Fig. 12(a) shows a slight asymmetry. For this symmetric antenna structure, the asymmetry arises because, during measurement, one port is terminated with 50 Ω and the plane wave is re-radiated through the opposite antenna. Even when a resonator is placed between the antennas, the improvement in $|S_{21}|$ through resonance has a practical limit; therefore, a symmetric antenna structure can still exhibit an asymmetric radiation pattern. Figures 13(a) and 13(b) show the simulated radiation patterns with and without the resonator. These results confirm that the resonator structure suppresses electrical mutual interference between the antennas without affecting the antenna gain or radiation pattern.



(a) $\phi = 0°$ (xz-plane)



(b) $\phi = 90°$ (yz-plane)

Fig. 13. Simulation results of the antenna radiation pattern according to the presence or absence of the resonator.

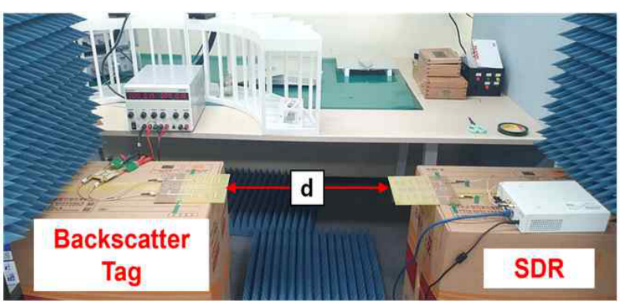


Fig. 14. Backscatter-communication experiment setup.

III. Communication-System Implementation and Range Measurement

The performance of the designed active tag was verified using GNU Radio and a USRP N200 SDR from NI. Digital data of "000110111" were transmitted on a 2.4 GHz carrier signal using on-off keying (OOK), and the magnitude of the signal received by the SDR was measured as a function of the distance between the active tag and the SDR. The data rate was 20 kbps. The transmit power of the SDR was 15 dBm (EIRP: 22 dBm). The experimental setup for wireless signal transmission and reception is shown in Fig. 14. With and without the high-isolation antenna, the distance was reduced incrementally from the maximum distance at which amplified reflection was clearly observed, and signal amplification was then evaluated at a point where the received signal was not saturated.

The received signal magnitudes were compared after nor-

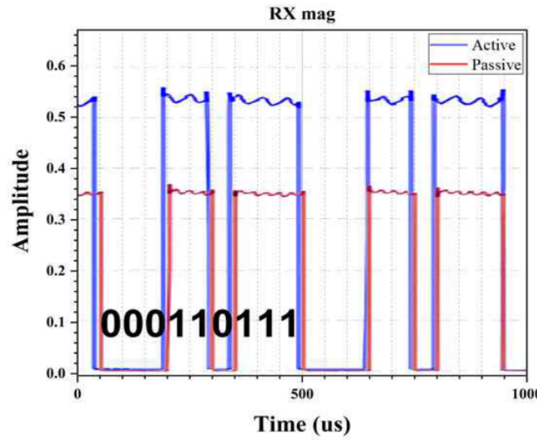


Fig. 15. Comparison of the normalized received Rx signal strength of active and passive tags ($d = 70$ cm).

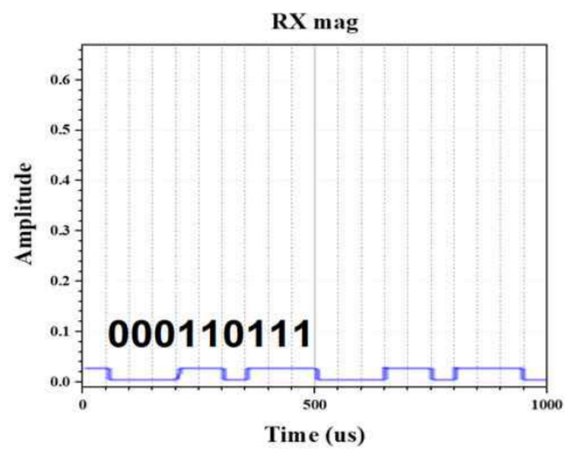


Fig. 16. Normalized received Rx signal strength with an active tag ($d = 2.5$ m).

malization with respect to the magnitude of the signal transmitted by the transmitter. When the communication distance (d) between the SDR and the tag was 70 cm, the normalized received backscattered signal magnitude was 0.36 for the passive tag without an amplifier. In contrast, when the active tag was used, the normalized magnitude increased to 0.55, corresponding to an increase of approximately 1.53 times (Fig. 15). When the communication distance of the active tag was 50 cm or less, the signal received by the SDR was observed to saturate. At a distance of 2.5 m, the normalized received signal magnitude was measured as 0.04 (Fig. 16). Beyond this distance, the signal strength became very small, and the signal characteristics could not be clearly distinguished.

Fig. 17 shows the waveform obtained at the point where the normalized backscattered signal magnitude became 0.36 while the communication distance was increased using the active tag. The results show that the active tag increased the communication distance by more than 40 cm compared with the passive tag. In addition, measurements confirmed that using the high-isolation antenna increased the communication distance by approximately 1.75 times compared with the case without the high-isolation antenna, corresponding to a 30 cm increase in communication range.

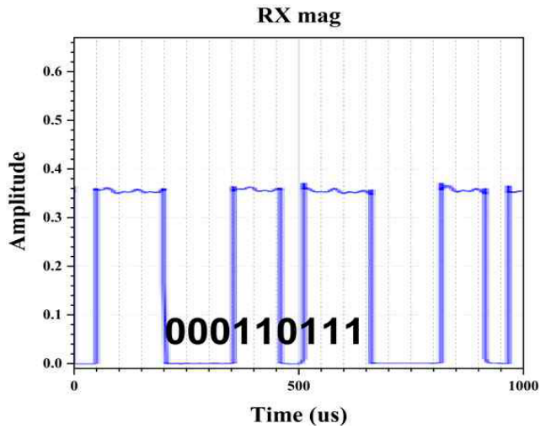


Fig. 17. Normalized received Rx signal strength with an active tag ($d = 110$ cm).

Table 3. Performance summary and comparison between backscatter systems.

Ref.	Comm. distance (m)	Data rate (kbps)	Frequency
This work	2.5	20	2.4 GHz
[3]	0.76	1	539 MHz
[7]	5	2.5	98.5 MHz
[8]	3	9.4	915 MHz
[9]	2.1	1	2.4 GHz

IV. Conclusion

In this study, an active tag was used to improve the communication range of backscatter communication, and experiments demonstrated that the communication range can be improved by using a reflection amplifier. As the distance decreased, the amplification ratio of the backscattered signal increased, and the experiments confirmed that the magnitude of a weak signal could be increased by approximately 1.5 times. When the active tag was used, the communication distance increased by 40 cm compared with that of the passive tag. When the high-isolation antenna was not used, the transmit power of the SDR was lower, and the communication distance decreased by 30 cm. Because backscatter communication re-radiates the received signal, the results confirm that a reduction in SDR transmit power shortens the communication distance; consequently, a high-isolation antenna can help increase the communication range.

If the gain of the reflection amplifier is increased further and the directivity of the antenna is improved, the communication range is expected to increase further, enabling communication over longer distances. As shown in Table 3, compared with previous backscatter-communication studies that transmit and receive digital signals, the proposed system achieves a high data rate in the 2.4 GHz band while maintaining a competitive communication range. This study experimentally demonstrates that, when an amplified-reflection active tag is used in indoor short-range ultralow-power communication such as IoT networks, communication performance can be significantly enhanced, enabling more precise communication.

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